

Exploration of Device-Free Sensing Ability Based on Bluetooth 6.0 Channel Sounding

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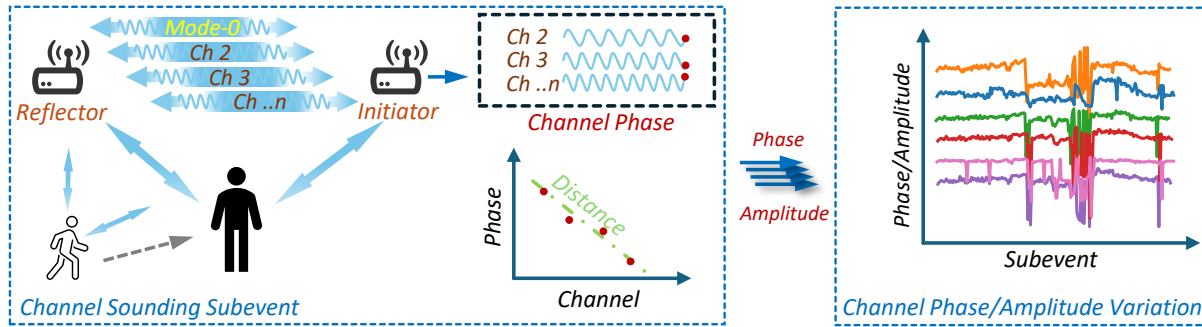


Figure 1: Overview of the proposed device-free human activity sensing method based on BLE 6.0 channel sounding.

Abstract

Bluetooth Low Energy (BLE) is pervasive in daily life, with its deployment scale and energy efficiency making it suitable for ubiquitous sensing applications. BLE 5.1 introduced Angle of Arrival/Departure (AoA/AoD) for direction-finding, enabling BLE devices to access phase and amplitude information similar to Channel State Information (CSI) in Wi-Fi sensing. In version 6.0, the new Channel Sounding (CS) feature enhances this capability through synchronization techniques that align the frequency and timing of two devices. This paper explores the sensing potential of CS-enabled BLE devices through practical experiments, presenting the first

demonstration of CS-based sensing. Compared to Wi-Fi-based sensing, the proposed technique achieves lower design costs and simpler deployment, enabling broader adoption in Internet of Things (IoT) applications. Experiments validate the synchronization advantages of BLE CS, enabling accurate distance measurements and demonstrating potential for human activity sensing. These results position BLE CS-based sensing as a promising direction for next-generation IoT technologies.

CCS Concepts

• **Human-centered computing** → **Ubiquitous computing**; *Mobile computing*.

Keywords

Bluetooth Low Energy (BLE); Channel Sounding (CS); Device-Free Sensing; Phase-based Ranging (PBR)

ACM Reference Format:

Cheng Wang, Manjiang Cao, Zijun Gong, and Tengfei Chang. 2025. Exploration of Device-Free Sensing Ability Based on Bluetooth 6.0 Channel Sounding. In *Companion of the the 2025 ACM International Joint Conference on Pervasive and Ubiquitous Computing (UbiComp Companion '25)*, October 12–16, 2025, Espoo, Finland. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3714394.3754369>

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UbiComp Companion '25, Espoo, Finland

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ACM ISBN 979-8-4007-1477-1/2025/10
<https://doi.org/10.1145/3714394.3754369>

1 Introduction

The rapid development of the Internet of Things (IoT) has driven the adoption of various technologies for human activity sensing, enabling applications such as health monitoring and smart home control. Among these, device-free sensing eliminates the need for wearable devices, improving usability and expanding application scenarios. Wireless signal-based sensing is particularly promising due to its non-invasive, privacy-preserving nature and robustness to lighting conditions, making it an effective alternative to vision-based solutions. Leveraging existing infrastructure such as Wi-Fi and BLE, these systems enable scalable and cost-effective device-free sensing.

Wi-Fi sensing systems can detect fine-grained activities, such as gestures [6], respiration [13, 15], and fall detection [10], through Channel State Information (CSI). However, CSI extraction is not standardized and requires specific hardware, limiting its scalability [16]. Additionally, accurate distance estimation remains a challenge, restricting its application to detecting relative movements [9].

Compared to Wi-Fi, BLE is more widely deployed and cost-effective. Since Bluetooth 5.1, it has introduced Angle of Arrival/Departure (AoA/AoD) [11], equipping BLE-based devices with I/Q sample measurement capabilities, similar to the CSI available in Wi-Fi. The newly introduced Channel Sounding (CS) feature in Bluetooth 6.0 further enhances the capability of BLE-based devices by enabling fine-ranging through phase difference measurements across multiple channels [12].

This paper investigates the sensing capability of CS-enabled BLE devices and demonstrates its feasibility through practical experiments (Fig. 1). Compared to Wi-Fi sensing, the proposed BLE-based approach achieves lower design costs and simpler deployment, enabling broader adoption in IoT applications.

The rest of this paper is organized as follows: Sec. 2 reviews device-free sensing based on Wi-Fi and BLE. Sec. 3 introduces BLE 6.0 CS protocol details and features. Sec. 4 explores the feasibility of BLE 6.0 CS for device-free sensing. Sec. 5 concludes the paper and discusses future work.

2 Wi-Fi Sensing and BLE's Catch-Up

In the Wi-Fi physical layer, the Long Training Field (LTF) in the packet preamble provides known subcarriers, allowing the receiver to estimate changes in channel phase and amplitude [5, 13], which is known as CSI. However, CSI is not standardized for sensing and only a limited set of network interface cards expose this low-level data [16]. Early studies primarily utilize amplitude variations in CSI, as reliable phase interpretation is restricted by synchronization noise, i.e., the carrier frequency offset and sampling time offset [5]. When two devices are synchronized, the phase of subcarriers should present a linear relationship with frequency, where the slope encodes path length. The synchronization noise usually distorts this relationship, resulting in inconsistent intercepts or non-linear trends [3]. Consequently, direct ranging is unreliable with Wi-Fi techniques, which limits the sensing ability to tracking relative changes, such as phase differences across antennas [8, 9, 15].

Along the development of Wi-Fi sensing techniques, new features are continuously integrated into Bluetooth specifications, which expands the sensing ability of BLE devices. In Bluetooth

5.1 [11], AoA/AoD is introduced to support the direction finding feature. It measures the phase difference over pairs of antennas to calculate the angle of the incoming signal [4, 7]. This feature requires the chip to expose the phase information from the I/Q samples, which enables the sensing ability, like the CSI used in Wi-Fi sensing. Bluetooth 5.1 does not provide phase-based ranging directly. Multi-carrier phase difference (MCPD) methods [2, 14] enable distance or multipath estimation based on BLE AoA/AoD, yet synchronization remains a challenge.

In Bluetooth 6.0 [1, 12], CS is introduced. Besides providing phase and amplitude measurement, CS also includes a Phase-Based Ranging (PBR) technique, which relies on hardware-level synchronization of frequency and timing. In addition to improved ranging accuracy, PBR also reduces hardware complexity, as it requires only a single antenna—unlike Bluetooth 5.1. With the rich features supported by CS, performing activity sensing through BLE devices is becoming increasingly promising, highlighting its advantages in IoT applications that emphasize low cost and simple design.

3 Channel Sounding and PBR in Bluetooth 6.0

This section first introduces the Channel Sounding protocol in Bluetooth 6.0. It then explains the theory behind how Phase-Based Ranging (PBR) works. Finally, the available physical layer features are shown.

3.1 Protocol Overview

The CS protocol organizes ranging operations into a series of procedures which comprises activities categories in 3 different hierarchy: events and subevents and steps, as shown in Fig. 2. Event is a logic unit defining the duration or number of subevents to conduct measurement. A subevent consists of multiple bilateral exchange between the two devices, an initiator and a reflector, to perform the measurements across different frequencies. The numbers steps within a subevents is defined as the number of bilateral packet exchanging.

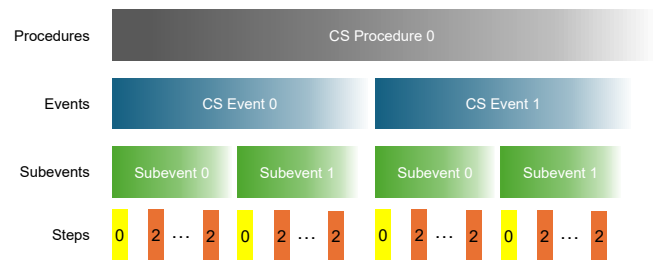


Figure 2: An example structure of channel sounding procedure. The numbers on the steps indicate the type, where 2 represents mode-2.

Steps have an associated mode which is numbered from 0 to 3 inclusive. Mode-0 step is responsible for establishing frequency and timing synchronization between devices. Mode-1 step is used to measure the Round-Trip-Time (RTT). Mode-2 step is used to measure the phase rotation of the RF signal between devices for PBR. Mode-3 step is used to measure both RTT and PBR.

A subevent always starts with a mode-0 step to calibrate the frequency and timing offset between the two devices. A sequence of non-mode-0 steps is followed after within the subevent. To the experiments conducted in this paper, multiple mode-2 steps for PBR are applied. To simplicity, only one CS event and one subevent is configured under a CS procedure.

3.2 Phase-Based Ranging

To perform PBR in mode-2 step, the initiator first transmits a continuous tone (CS tone) on a specific channel. After a brief idle period, the reflector transmits its own CS tone on the same channel. Both parties will measure the phase of the signals emitted by each other and send the measurement results to the other party. According to the protocol, the final phase shift calculation will be performed by the initiator and reported to the application layer. By repeating this over multiple channels, i.e. multiple mode-2 steps, it calculates the phase slope across frequencies—used to estimate distance.

When the continuous tone travels from one device to another, the signal experiences a phase shift that depends on the distance and wavelength. The theory behind it can be expressed as Eq. 1, where $\phi(f)$ is the phase shift at frequency f , x is the distance between the two devices and c is the speed of light.

$$\phi(f) = -2\pi f \frac{2x}{c} \bmod 2\pi \quad (1)$$

Because phase wraps around every 2π , a single frequency gives ambiguous distance due to phase wrapping. To resolve this, PBR transmits tones over multiple frequencies and calculates the phase differences between them, as shown in Eq. 2.

$$\Delta\phi = \phi(f_2) - \phi(f_1) \quad (2)$$

This phase slope across frequencies is proportional to the distance as shown in Eq. 3.

$$x = \frac{c}{4\pi} \frac{\Delta\phi}{\Delta f} \quad (3)$$

3.3 Measurable Physical Layer Features

As a pre-step to explore the sensing ability of channel sounding, we implement it with two nRF54L15¹ development boards which supports the latest CS functionalities. Then we designed experiments to extract the raw information reported by each CS subevent.

The implementation is based on the latest SDK (2.9.1) provided from Nordic community. The two boards are configured as initiator and reflector, as explained in CS procedure in Sec. 3. 16 channels are configured to use for the PBR distance measurements. A channel width of 1 MHz rather than the usual 2 MHz is used to avoid distance ambiguity. As a result, the number of available BLE CS channels increased from 40 to 72, excluding channels for advertising.

At the end of each subevent, the application layer received the phase measurement results for all enabled channels, stored in I/Q format. For PBR measurements, each channel contained two sets of I/Q data, referred to as “local” and “peer” in the SDK (Fig. 3 (a)).

To obtain the true phase changes of the signal (*signal*), we compute the difference between the *peer* and *local* phases.

Since the phase varies cyclically within $[-\pi, \pi]$, any difference between two phase values exceeding one cycle is considered a phase jump. To obtain a continuous phase change, we perform phase unwrapping on the *signal* phase values (Fig. 3 (c)). Before unwrapping, the phase values exhibit periodic variation (Fig. 3 (b)).

For each CS subevent, we can extract the bidirectional phase difference *signal* across all channels during signal propagation. The slope of the linear fit of these phase values represents the distance information. Since synchronization is already achieved via **mode-0**, under ideal external context, extracting the channel phase changes that represent the propagation path distance does not require additional signal processing steps.

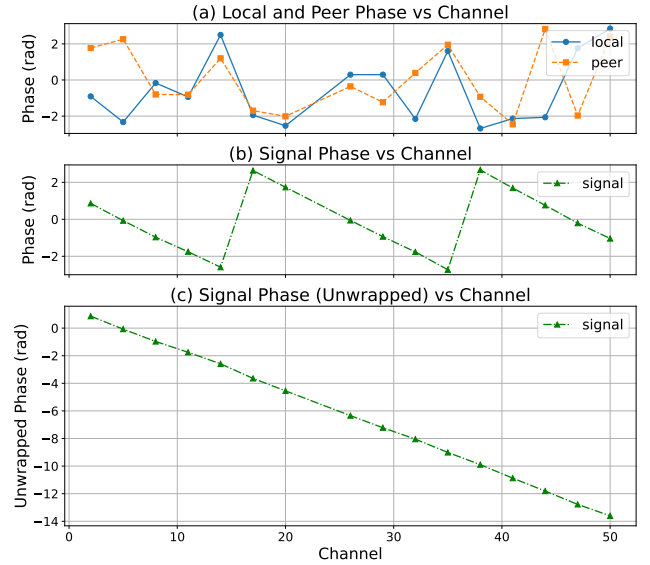


Figure 3: Phase information for each channel from a single channel sounding subevent. (a) shows the “local” and “peer” phases; their difference gives the true signal phase change (b). After unwrapping, the signal phase across channels (c) forms an approximately straight line.

4 Exploration of Device-Free Sensing with CS

Human activities influence signal propagation primarily by modifying the multipath environment, resulting in changes to the amplitude and phase of the received signal. These variations can be extracted from the I/Q data reported by subevents. In this preliminary work, we present both the amplitude and phase of the signal. We validate the sensing feasibility by two questions:

- Does the phase/amplitude information remain stable in different stable environments?
- Does the phase/amplitude information react consistently to the human activities?

4.1 Stability Across Different Contexts

To answer the first question, we conduct experiments by connecting the antennas of two boards through two extended RF cables with 2

¹<https://www.nordicsemi.com/Products/nRF54L15>

and 3 meters long. From Fig. 4, it can be observed that with RF cables of different lengths, the phase across all channels remains stable (Fig. 4 (a)). The major phase variation is within ± 0.05 radians (Fig. 4 (b)). Both amplitude and phase exhibit similar behavior. However, it is noteworthy that despite setting equal transmission power for both sides in the program, there are still some differences in reality (Fig. 4 (c)).

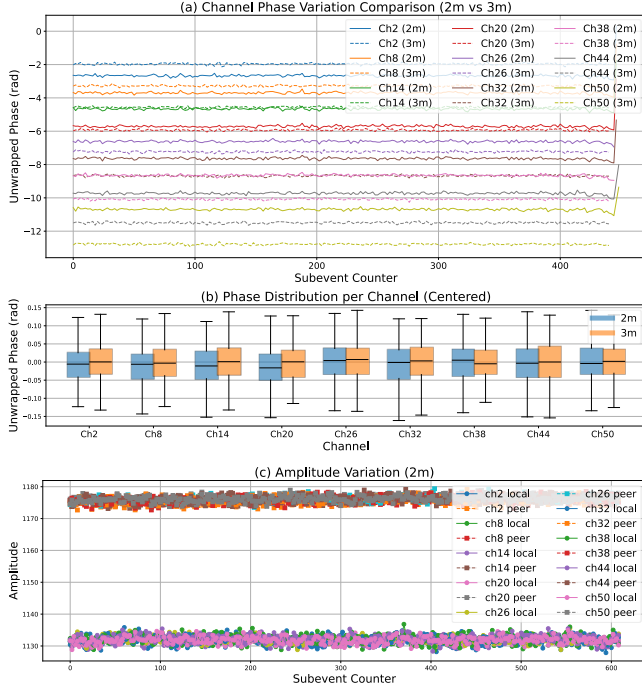


Figure 4: The amplitude and phase of received signals for 9 out of 16 channels at distances of 2m and 3m. (a) shows the phase variation; (b) quantifies the phase variation, with the mean for each channel normalized to 0; (c) illustrates the amplitude differences between transmitted and received signals at the same distances.

4.2 Reacting to Human Activity

To address the second question, we placed two development boards indoors, spaced 1.2 meters apart and 70 centimeters above the ground. A person stood near the line of sight (LOS) and remained at three different positions for 20 seconds each. As shown in Fig. 5, both the amplitude and phase of the signal can reflect the multipath changes caused by the presence of a human body. Due to page limitations, only the local amplitude is plotted. Even without any signal processing techniques, we can clearly observe three distinct patterns.

5 Conclusion and Future Work

This paper presents an experimental study on device-free sensing using the CS feature introduced in Bluetooth 6.0. With the implementation over two CS enabled boards, we demonstrate the sensing feasibility through verifying the stability of phase information over

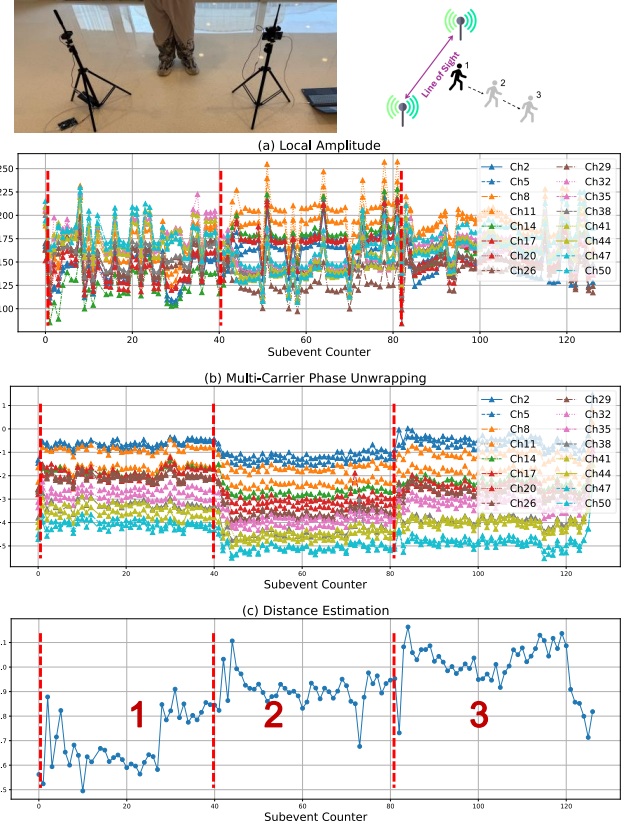


Figure 5: Amplitude and phase variations of the received signal at three different positions. The amplitude shows only the signal transmitted from one side (local), while the phase reflects the actual phase changes of the signal.

time at difference distance, and the consistent reactions to human activities. The CS-based sensing enables a low-cost design and easy deployment IoT solution, comparing to Wi-Fi based sensing, at a even higher deployment scale. A complete sensing solution with CS enabled BLE devices is to be studied in the future. The expecting works include investigating the activities resolutions of based on the phase/amplitude information over time, and the exploration of the potential of more signal processing method used for sensing.

Acknowledgments

This work is funded by the Guangdong Provincial Key Lab of Integrated Communication, Sensing and Computation for Ubiquitous Internet of Things, with Grant numbers 2023B1212010007, Guangzhou Municipal Science and Technology Project 2023A03J0011 and the Guangzhou Municipal Key Laboratory on Future Networked Systems 2024A03J0623.

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